

A quadratic-cost backward-error solver for general real systems

An affirmative solution to IE-12 in the stated exact-real model

Sidney Holden

Center for Computational Biology, Flatiron Institute, Simons Foundation

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Abstract

We give an explicit randomized algorithm for the normalized, fully accessible real linear-system problem in IE-12. Its deterministic worst-case operation count is $O(n^2\varepsilon^{-3})$; with probability greater than 0.997 its nonzero output satisfies $\|Ax - b\|_2 / \|x\|_2 \leq 5\varepsilon/8$. Thus the answer to IE-12 is affirmative with $q = 3$. The algorithm rounds A once to a nearby scaled integer matrix. A weighted-pattern multiplication scheme supports repeated products with this matrix and its transpose more cheaply than dense products. A rectangular augmented system supplies an exact kernel, and a fixed-length randomly started polynomial filter finds an approximate kernel vector. A clamping step converts it to the required backward-error guarantee without perturbing b . The logarithm in the iteration count is canceled by the preprocessed product cost, not removed from convergence itself. All essential lemmas, probability estimates, model details, and cost bounds are proved below.

Verification status. This is a complete mathematical argument with executable exact-arithmetic component tests and floating-point illustrations. The submitted draft was prepared with ChatGPT assistance; authorship is recorded as Sidney Holden at his request. Independent informal Codex AI-agent review is documented in the accompanying submission record. This is not external human peer review or proof-assistant verification. The theorem is about exact-real scalar complexity, not finite-precision stability, bit complexity, or practical speed.

1 Problem, result, and mechanism

IE-12 asks for absolute constants $C, q > 0$ and an algorithm that, given a nonsingular $A \in \mathbb{R}^{n \times n}$ with promised $\|A\|_2 = 1$, a nonzero $b \in \mathbb{R}^n$, and $0 < \varepsilon < 1/2$, uses at most $Cn^2\varepsilon^{-q}$ operations and returns a nonzero vector satisfying

$$\mathbb{P}\left\{\frac{\|Ax - b\|_2}{\|A\|_2 \|x\|_2} \leq \varepsilon\right\} \geq 0.99. \quad (1)$$

The allowed primitives are exact-real scalar arithmetic, square roots, comparisons, and independent standard Gaussian or random-bit draws. Input entries are accessible; normalization need not be computed. The metric allows perturbations in A only [1].

The cited general-system result has cost $O(n^2 \log(n/\delta)/\varepsilon)$ for failure probability δ ; the corresponding PSD result is $O(n^2/\sqrt{\varepsilon})$ [2, Corollaries 17 and 25]. Our construction addresses the stated operation-count question by exploiting entry access and preprocessing. It does not assert dimension-independent convergence of a black-box Krylov iteration.

Theorem 1 (Affirmative answer to IE-12). *There is an explicit randomized algorithm in the stated exact-real model and an absolute constant C such that, for every $n \geq 1$, every real $A \in \mathbb{R}^{n \times n}$ with $\|A\|_2 = 1$, every $b \neq 0$, and every $0 < \varepsilon < 1/2$, the algorithm always returns $x \neq 0$, uses at most $Cn^2\varepsilon^{-3}$ operations and $O(n^2)$ real-scalar storage, and satisfies*

$$\mathbb{P}\left\{\frac{\|Ax - b\|_2}{\|x\|_2} \leq \frac{5\varepsilon}{8}\right\} > 0.997. \quad (2)$$

Its output also always satisfies $\|x\|_2 \leq 8\|b\|_2/\varepsilon$. The constant is independent of all input data and of conditioning. Nonsingularity is not required for this stronger statement.

The matrix-product construction is related to the Mailman method, which saves a logarithmic factor for preprocessed constant-alphabet matrices [3]. Here the rounded matrix has an alphabet that can grow with n

and $1/\varepsilon$, so applying that result as a constant-alphabet black box would not suffice. We instead charge an integer coefficient a a weight $1 + |a|$, enumerate only patterns of bounded total weight, and process heavy entries individually. The resulting estimates are

$$W = O(n^2/\varepsilon), \quad (3)$$

$$\text{one product} = O(n4^k + W/k + n), \quad (4)$$

$$\text{number of iterations} = O(k/\varepsilon^2), \quad (5)$$

where $k = \max\{1, \lfloor \log_{16} n \rfloor\}$. Since $k4^k = O(n)$, multiplying (4) by (5) gives $O(n^2/\varepsilon^3)$. Both preprocessing and every table computation are included.

Throughout, vector and matrix norms without a subscript other than F are Euclidean and spectral norms, respectively. Transpose is written $\top$. Indexed scalar-array reads and writes can be counted at unit cost without changing any bound. Equivalently, preprocessing builds a fixed-wiring scalar arithmetic circuit for each subsequent product. No floor, logarithm, hashing, SVD, or real-number digit-extraction oracle is assumed.

2 One bounded-cost rounding of the input

Set

$$\beta = \|b\|_2, \quad c = b/\beta, \quad \rho = \eta = \varepsilon/8, \quad h = \frac{\varepsilon}{64\sqrt{n}} = \frac{\rho}{8\sqrt{n}}. \quad (6)$$

For each $t = A_{ij}/h$, let $m = \lfloor t \rfloor$ and $p = t - m$. Independently set $Z_{ij} = m + 1$ with probability p and $Z_{ij} = m$ otherwise. Write $Q = hZ$.

2.1 Implementation using exactly the permitted primitives

The floor is not taken as a primitive. Starting with $m = 0$, if $t \geq 0$, increment m until $m + 1 > t$; if $t < 0$, decrement m until $m \leq t$. This costs $O(1 + |t|)$ scalar additions and comparisons and gives the exact floor.

An exact uniform random variable can be obtained in constant cost from three independent standard Gaussians. Put

$$R = \sqrt{G_1^2 + G_2^2 + G_3^2}, \quad U = \frac{1 + G_1/R}{2} \quad (R > 0). \quad (7)$$

If $R = 0$, assign $U = 1/2$ without retrying. This event has probability zero. The normalized Gaussian vector is uniform on the unit sphere in $\mathbb{R}^3$, by rotational invariance. A strip whose first coordinate lies in $[z, z + dz]$ has area $2\pi dz$, whereas the sphere has area 4π . Consequently G_1/R is uniform on $[-1, 1]$, and U is uniform on $[0, 1]$. The comparison $U < p$ therefore implements the rounding rule exactly. Independent triples are used for distinct entries, and the later starting vector uses fresh draws. There is no rejection loop and no expected-time qualification.

Lemma 2 (Weight, cost, and spectral error). *For every realization of the rounding,*

$$W := \sum_{i,j} (1 + |Z_{ij}|) \leq (2 + 64/\varepsilon)n^2 \leq 65n^2/\varepsilon. \quad (8)$$

The entire rounding costs $O(n^2/\varepsilon)$ operations, including computation of the integer parts. Moreover,

$$\mathbb{P}\{\|Q - A\|_2 > \rho\} \leq 2 \exp((2 \log 9 - 32)n) < 10^{-3}. \quad (9)$$

Proof. Since $|Z_{ij}| \leq |A_{ij}|/h + 1$, Cauchy–Schwarz gives

$$\sum_{i,j} |A_{ij}| \leq n\|A\|_F \leq n\sqrt{n}\|A\|_2 = n^{3/2}.$$

Thus $W \leq 2n^2 + n^{3/2}/h = (2 + 64/\varepsilon)n^2$. Because $2\varepsilon < 1$, this implies (8). The comparison-floor costs sum to $O(n^2 + \sum |A_{ij}|/h) = O(n^2/\varepsilon)$; all other entrywise work is constant per entry. These estimates are deterministic.

Let $E = Q - A$. Its entries are independent, centered random variables, each supported in an interval of length at most h . We include the elementary exponential estimate used here. If Y is centered and supported in an interval of length L , then, for $K(s) = \log \mathbb{E}e^{sY}$, the second derivative is the variance under an exponentially tilted distribution. Any variable in such an interval has variance at most $L^2/4$: subtract the midpoint and use $\text{Var}(Y) \leq \mathbb{E}(Y - \text{midpoint})^2$. Hence $K''(s) \leq L^2/4$ and $K(0) = K'(0) = 0$, giving

$$\mathbb{E}e^{sY} \leq e^{s^2 L^2/8}.$$

For fixed unit vectors $u, v \in \mathbb{R}^n$, independence and $\sum_{ij} u_i^2 v_j^2 = 1$ imply

$$\mathbb{E}e^{su^\top E v} \leq e^{s^2 h^2/8}.$$

Optimizing the exponential Markov bound in s and applying it to both signs therefore yields

$$\mathbb{P}\{|u^\top E v| > t\} \leq 2e^{-2t^2/h^2}. \quad (10)$$

There is a $1/4$ -net $\mathcal{N}$ of the unit sphere with $|\mathcal{N}| \leq 9^n$. Indeed, take a maximal $1/4$ -separated set; its disjoint radius- $1/8$ balls fit in the radius- $9/8$ ball, so volume comparison proves the cardinality bound, and maximality proves the net property. For every matrix E , approximation of both unit vectors in a maximizing bilinear form gives

$$\|E\|_2 \leq 2 \max_{u,v \in \mathcal{N}} |u^\top E v|.$$

The two approximation errors are each at most $\|E\|_2/4$. A union bound in (10), with $t = \rho/2$, proves

$$\mathbb{P}\{\|E\|_2 > \rho\} \leq 2 \cdot 9^{2n} \exp\left(-\frac{\rho^2}{2h^2}\right) = 2 \exp((2 \log 9 - 32)n).$$

Since $2 \log 9 < 5$, the last quantity is at most $2e^{-27} < 10^{-3}$. The nets and exponential functions are proof devices, not computations performed by the algorithm. $\square$

3 Weighted-pattern products with an integer matrix

We next prove the product construction independently of the random rounding. An integer pattern is a finite sequence $s = (a_1, \dots, a_m)$, with

$$\omega(s) = \sum_{r=1}^m (1 + |a_r|).$$

The empty pattern has weight zero. Let $\mathcal{P}_k$ be the collection of all patterns of weight at most an integer $k \geq 1$.

Lemma 3 (A small catalogue with explicit identifiers). *The catalogue satisfies $|\mathcal{P}_k| \leq 4^k$. It can be enumerated, together with a distinct identifier for each pattern and the head and tail identifiers of every nonempty pattern, in $O(k4^k)$ scalar operations and storage. Each given pattern in $\mathcal{P}_k$ can be encoded in $O(k)$ operations.*

Proof. Encode an integer coefficient with symbols in $\{0, 1, 2\}$ as follows: 0 is the single symbol 0 ; $+r$ is r copies of 1 followed by 0 ; $-r$ is r copies of 2 followed by 0 . Concatenate the coefficient codes and pad the resulting word by symbols 3 to total length k . The unpadding length is exactly the weight. Zeros terminate coefficients; the first 3 , if present, starts the padding. Thus the encoding is injective. There are 4^k words of length k over four symbols.

The identifier is the base-four value of the padded word, computed by the recurrence code $\leftarrow 4 \text{ code} + \text{symbol}$. Leading zeros cause no ambiguity because every padded word has length k . No bit-packing or digit extraction is used. To enumerate the catalogue, recursively append all integers of weight at most the remaining budget. There are at most 4^k visited patterns, at most $O(k)$ work at each visit, and at most k coefficients to store per pattern. Computing the tail code afresh costs another $O(k)$ per pattern, still within $O(k4^k)$. The power 4^k itself is formed by repeated multiplication. $\square$

Lemma 4 (Preprocessed integer products). *Let $Z \in \mathbb{Z}^{n \times n}$, let $1 \leq k \leq n$, and put $W = \sum_{ij} (1 + |Z_{ij}|)$. In $O(W + k4^k)$ preprocessing operations one can construct exact product routines for both Zv and $Z^\top v$, valid for arbitrary real v . Each product costs*

$$O(n4^k + W/k + n) \quad (11)$$

operations. The storage, including input, descriptors, and a product's intermediate table, is $O(n^2 + k4^k + n4^k)$ real-scalar words.

Proof. Packing the rows. Give an entry a weight $1 + |a|$. If this exceeds k , mark it heavy and store it as a standalone multiplication. In each contiguous run of remaining light entries, greedily pack entries into maximal consecutive bins whose total weight is at most k . Each bin is a catalogue pattern, recorded by its identifier and starting column. Its length is at most k .

Let H be the number of heavy entries, and let W_L, W_H be the total light and heavy weights. In a light run, any two consecutive bins have combined weight greater than k : otherwise the first entry of the second bin could have been added to the first. Pairing consecutive bins shows that the run has at most $2W_{\text{run}}/k + 1$ bins. There are at most $H + n$ nonempty light runs. Since $H \leq W_H/k$, the number S of all bins and standalone heavy terms satisfies

$$S \leq 2W_L/k + (H + n) + H \leq 2W/k + n. \quad (12)$$

Each term covers at least one original entry, so also $S \leq n^2$. Scanning costs $O(n^2)$, and encoding a bin costs $O(k)$, including padding. Thus the scan and encoding cost is $O(n^2 + Sk) = O(W + nk) = O(W)$ because $W \geq n^2$ and $k \leq n$. Heavy coefficients are never expanded in unary.

All shifted pattern products. For a queried vector v , build a table $F(j, s)$ for $j = n, n-1, \dots, 1$ and $s \in \mathcal{P}_k$. Initialize $F(n+1, s) = 0$ and $F(j, \emptyset) = 0$. If $s = (a, s')$ is nonempty, use

$$F(j, s) = av_j + F(j+1, s'). \quad (13)$$

Induction on j proves that $F(j, s)$ is the dot product of s with the segment of v starting at j , with zero padding beyond n . In particular it is the exact contribution of a stored bin. The head and tail identifiers were computed in preprocessing. Each table entry now costs at most one multiplication and one addition, not $O(k)$ operations. Initializing and filling a direct-address table of $(n+1)4^k$ slots costs $O(n4^k)$, including unused slots.

For each row, sum one looked-up value for each bin and one scalar product for each heavy term. This costs $O(S + n)$ operations. Together with (12), this proves (11). The table is recomputed for every queried vector; its entire cost has just been counted.

Apply the same preprocessing separately to $Z^\top$, using the same catalogue. Its weight is also W and transposing costs $O(n^2)$. Two routines change the constants only. At most n^2 terms are stored per routine; the catalogue requires $O(k4^k)$ words, and a table requires $O(n4^k)$ words. The two products can reuse table workspace, proving the storage claim. $\square$

Remark 5 (Meaning of table access). All identifiers are integers constructed by the explicitly charged loops and scalar arithmetic. Their range is $0, \dots, 4^k - 1$; ordinary indexed storage suffices. Alternatively, the identifiers tell a preprocessor which previously computed scalar to connect to each output sum. Evaluating the resulting straight-line circuit needs only scalar arithmetic. Neither a hash table nor a nonuniform advice string about A is required.

For $Q = hZ$, multiply a routine's output by h , at $O(n)$ additional cost. The same statement applies to $Q^\top$.

4 A condition-independent approximate-kernel filter

The next lemma deliberately uses a rectangular matrix. Its nontrivial kernel is guaranteed by dimension, not by a computed singular vector, a nonsingular rounded matrix, or consistency of a rounded linear system.

Lemma 6 (Fixed-time randomized kernel filtering). *Let $B \in \mathbb{R}^{(d-1) \times d}$ satisfy $\|B\|_2^2 \leq 4$, and let $\eta > 0$. Let ℓ be the least nonnegative integer with $2^\ell \geq 10^9 d$. Let T be the least nonnegative integer satisfying $T\eta^2 \geq 4\ell$. Draw $g \sim N(0, I_d)$ and compute*

$$z_0 = g, \quad z_{t+1} = z_t - \frac{1}{4} B^\top B z_t, \quad 0 \leq t < T. \quad (14)$$

With probability greater than 0.998, $z_T \neq 0$ and

$$\|B z_T\|_2 \leq \eta \|z_T\|_2. \quad (15)$$

No condition number or spectral gap enters T .

Proof. There exists a unit vector $v \in \ker B$; fix one for analysis only. Put $a = |v^\top g|$ and $G = \|g\|_2$. A standard Gaussian has density at most $1/\sqrt{2\pi}$, so

$$\mathbb{P}\{a < 10^{-3}\} \leq \sqrt{2/\pi} 10^{-3} < 10^{-3}.$$

Also $\mathbb{E}G^2 = d$, hence Markov's inequality gives $\mathbb{P}\{G^2 > 1000d\} \leq 10^{-3}$. On their intersection, an event of probability greater than 0.998,

$$a \geq 10^{-3}, \quad G^2/a^2 \leq 10^9 d. \quad (16)$$

Let $H = B^\top B$ and $M = I - H/4$. The eigenvalues λ of H lie in $[0, 4]$, so M is a positive semidefinite contraction and $Mv = v$. The iterate is $z_T = M^T g$, where T denotes an integer power here, not transpose. Thus $v^\top z_T = v^\top g$, and

$$\|z_T\|_2 \geq a > 0. \quad (17)$$

Use an orthonormal eigenbasis of H , with coordinates g_j and $z_{T,j} = (1 - \lambda_j/4)^T g_j$. The coordinates with $\lambda_j \leq \eta^2/2$ contribute at most $\frac{1}{2}\eta^2 \|z_T\|_2^2$ to $\|B z_T\|_2^2 = \sum_j \lambda_j z_{T,j}^2$.

For the remaining coordinates, the inequality $1 - s \leq e^{-s}$ for $0 \leq s \leq 1$ gives

$$\lambda(1 - \lambda/4)^{2T} \leq \lambda e^{-T\lambda/2}.$$

Because $T \geq 4/\eta^2$, the function $\lambda \mapsto \lambda e^{-T\lambda/2}$ is decreasing on $[\eta^2/2, \infty)$. Consequently the high-eigenvalue contribution is at most

$$\frac{\eta^2}{2} e^{-T\eta^2/4} G^2. \quad (18)$$

By construction, $\ell \geq \log_2(10^9 d) \geq \log(10^9 d)$ and $T\eta^2/4 \geq \ell$. Therefore (16) bounds (18) by $\eta^2 a^2/2$, which is at most $\eta^2 \|z_T\|_2^2/2$ by (17). Adding the two parts proves (15). This proof also shows why no additional $\log(1/\eta)$ factor is needed: the factor λ in the residual is retained when bounding the high-eigenvalue part. $\square$

For our application take $d = n + 1$ and

$$B = [Q \quad -c]. \quad (19)$$

On the rounding-success event of Lemma 2,

$$\|Q\|_2 \leq 1 + \rho < 17/16, \quad \|B\|_2^2 \leq \|Q\|_2^2 + \|c\|_2^2 \leq (17/16)^2 + 1 = 545/256 < 4. \quad (20)$$

For $z_t = (u_t, \alpha_t)$, a step of (14) is exactly

$$\begin{aligned} r_t &= Qu_t - c\alpha_t, \\ u_{t+1} &= u_t - \frac{1}{4} Q^\top r_t, \\ \alpha_{t+1} &= \alpha_t + \frac{1}{4} c^\top r_t. \end{aligned} \quad (21)$$

It uses one product with each of $Q, Q^\top$ and $O(n)$ further operations. In particular neither $B^\top B$ nor a dense Gram matrix is formed. The vector v in the proof is not computed.

5 Conversion to backward error with b unchanged

Lemma 7 (Clamped projective conversion). *Assume (6), $\|Q - A\|_2 \leq \rho$, and $z = (u, \alpha) \neq 0$ satisfies*

$$\|Qu - c\alpha\|_2 \leq \eta \|z\|_2. \quad (22)$$

Then $u \neq 0$. Define $\text{sign}_+(\alpha) = 1$ for $\alpha \geq 0$ and -1 otherwise, and put

$$\alpha' = \text{sign}_+(\alpha) \max\{|\alpha|, \eta \|u\|_2\}, \quad x = \frac{\beta}{\alpha'} u. \quad (23)$$

This vector is nonzero and satisfies

$$\frac{\|Ax - b\|_2}{\|x\|_2} \leq \rho + 4\eta = 5\varepsilon/8, \quad \|x\|_2 \leq \beta/\eta. \quad (24)$$

In particular the definition remains valid when $\alpha = 0$.

Proof. If $u = 0$, then $\alpha \neq 0$ and $\|Qu - c\alpha\|_2 = |\alpha| = \|z\|_2$, contradicting $\eta < 1$. Moreover,

$$|\alpha| = \|c\alpha\|_2 \leq \|Qu\|_2 + \|Qu - c\alpha\|_2 \leq (1 + \rho) \|u\|_2 + \eta \|z\|_2.$$

Using $\|z\|_2 \leq \|u\|_2 + |\alpha|$ gives

$$\frac{\|z\|_2}{\|u\|_2} \leq \frac{2 + \rho}{1 - \eta} < 3, \quad (25)$$

since $\rho = \eta < 1/16$. The clamping rule gives $|\alpha'| \geq \eta \|u\|_2 > 0$ and $|\alpha' - \alpha| \leq \eta \|u\|_2$, including when $\alpha = 0$. Thus $x \neq 0$ and $\|x\|_2 \leq \beta/\eta$. Since $b = \beta c$ exactly,

$$\begin{aligned} \frac{\|Ax - b\|_2}{\|x\|_2} &= \frac{\|Au - c\alpha'\|_2}{\|u\|_2} \\ &\leq \|A - Q\|_2 + \frac{\|Qu - c\alpha\|_2}{\|u\|_2} + \frac{|\alpha' - \alpha|}{\|u\|_2} \\ &\leq \rho + \eta \frac{\|z\|_2}{\|u\|_2} + \eta \leq \rho + 4\eta. \end{aligned}$$

This establishes the claimed guarantee for the original A and b . □

For completeness, the backward perturbation is fully explicit. Given any nonzero output x , set

$$\Delta A = \frac{(b - Ax)x^\top}{\|x\|_2^2}. \quad (26)$$

Then $(A + \Delta A)x = b$ and $\|\Delta A\|_2 = \|Ax - b\|_2 / \|x\|_2$. Conversely, $\Delta Ax = b - Ax$ implies $\|\Delta A\|_2 \geq \|Ax - b\|_2 / \|x\|_2$, so this is exactly the minimum spectral-norm perturbation, not merely an upper bound on a different metric. On success, (24) and $\|A\|_2 = 1$ give $\|\Delta A\|_2 / \|A\|_2 \leq 5\varepsilon/8$. No perturbation of b is needed.

6 Complete algorithm and deterministic accounting

The following is the exact-real algorithm. The shorthand for rounding and product preprocessing denotes precisely the loops proved above. The iteration count is implemented by a comparison, rather than a ceiling primitive.

Listing 1: Exact-real solver; all loops have deterministic cost bounds.

```

Input: entries of A, b; promised ||A||_2 = 1; b != 0; 0 < eps < 1/2
beta = sqrt(sum_i b[i]^2); c = b / beta
rho = eta = eps / 8; h = eps / (64 * sqrt(n))

For each entry A[i,j]:
  t = A[i,j] / h
  compute m = floor(t) by the comparison/increment loops
  draw three fresh independent standard Gaussians
  obtain U from (1 + G1 / sqrt(G1^2 + G2^2 + G3^2)) / 2
    using U = 1/2 when the square-root denominator is zero
  Z[i,j] = m + 1 if U < t - m, and m otherwise

k = 1; power = 16
while 16 * power <= n:
  power = 16 * power; k = k + 1
preprocess weighted-pattern products for Z and transpose(Z)
Q_product(v) = h * Z_product(v)
QT_product(v) = h * ZT_product(v)

ell = 0; power = 1
while power < 10^9 * (n + 1):
  power = 2 * power; ell = ell + 1
Draw n + 1 fresh independent standard Gaussians as (u, alpha)
t = 0
while t * eta^2 < 4 * ell:
  r = Q_product(u) - c * alpha
  u = u - QT_product(r) / 4
  alpha = alpha + dot(c, r) / 4
  t = t + 1

s = sqrt(dot(u,u))
if s == 0: return beta * e_1
alpha_prime = max(abs(alpha), eta * s)
if alpha < 0: alpha_prime = -alpha_prime
return (beta / alpha_prime) * u

```

6.1 A uniform bound valid even for small n

Lemma 8 (Cancellation parameters). *For the k, ℓ, T used in the algorithm,*

$$1 \leq k \leq n, \quad n < 16^{k+1}, \quad k4^k \leq 4n, \quad (27)$$

$$\ell \leq 4k + 35 \leq 39k, \quad T \leq 10000 k / \varepsilon^2. \quad (28)$$

They can be determined with the permitted arithmetic and comparisons.

Proof. For $n \geq 16$, $16^k \leq n < 16^{k+1}$. Since $k \leq 4^k$ for positive integers, $k4^k \leq 16^k \leq n$. For $1 \leq n < 16$, $k = 1$ and $k4^k = 4 \leq 4n$. The remaining assertions in (27) follow directly. Since $10^9 < 2^{30}$ and $n + 1 \leq 2n < 2 \cdot 16^{k+1}$,

$$10^9(n + 1) < 2^{4k+35},$$

so the least possible ℓ is at most $4k + 35 \leq 39k$. The least T with $T\eta^2 \geq 4\ell$ obeys

$$T \leq \frac{4\ell}{\eta^2} + 1 = \frac{256\ell}{\varepsilon^2} + 1 \leq \frac{9984k}{\varepsilon^2} + 1 \leq \frac{10000k}{\varepsilon^2}.$$

Repeated multiplication and comparison implement the k and ℓ loops; the iteration loop realizes exactly this T . Logarithms, floors, and ceilings in their mathematical descriptions are not unit-cost operations in the algorithm. $\square$

Proof of Theorem 1. Correctness and probability. By Lemma 2, the rounding-success event $\mathcal{E} = \{\|Q - A\|_2 \leq \rho\}$ has probability greater than 0.999. Conditional on any realized Q in this event, (20) holds, and the fresh Gaussian start permits application of Lemma 6. With conditional probability greater than 0.998 the computed vector is nonzero and satisfies (22). Lemma 7 then gives the claimed $5\varepsilon/8$ error for the original input. A union bound bounds the total failure probability by a quantity strictly less than $0.001 + 0.002 = 0.003$. The conditional argument does not require independence of the success events; only the fresh starting draw is needed.

All realizations are well-defined. The Gaussian-to-uniform construction has an explicit branch at its zero denominator. Also $\beta > 0$, $h > 0$, and no division takes place inside the filter except by 4. If $\|u\|_2 = 0$ at the end, the fallback βe_1 is nonzero. Otherwise the clamped denominator is nonzero. Consequently every realization, including those outside the success event, returns a nonzero vector after the same bounded number of iterations. Clamping gives $\|x\|_2 \leq \beta/\eta$; the fallback also satisfies this bound because $\eta < 1$. There is no condition-number-sensitive stopping rule.

Operation count. Rounding costs $O(n^2/\varepsilon)$. By Lemma 4, both product routines cost $O(W + k4^k)$ to preprocess. One filter step costs $O(n4^k + W/k + n)$, including both product tables and all updates. Normalization, loop setup, and final conversion cost at most $O(n + k + \ell)$ in addition. The iteration comparison and counter cost $O(T)$ in total and are absorbed by the step bound. Therefore the entire cost is at most

$$\begin{aligned} O(n^2/\varepsilon) + O(W + k4^k) + O\left(\frac{k}{\varepsilon^2} \left(n4^k + \frac{W}{k} + n\right)\right) \\ = O(n^2/\varepsilon) + O\left(\frac{kn4^k + W + nk}{\varepsilon^2}\right) = O(n^2/\varepsilon^3). \end{aligned} \quad (29)$$

The last equality uses $k4^k \leq 4n$, $k \leq n$, and $W \leq 65n^2/\varepsilon$. Every estimate is uniform in all inputs and all random outcomes. This proves existence of an absolute operation-count constant C with $q = 3$.

Storage. The input and two term lists require $O(n^2)$ words. By (27), $k4^k = O(n)$ and $n4^k = O(n^2)$, so the catalogue and reusable product tables fit in $O(n^2)$ words. The vectors, counters, and temporary scalars need less space. This proves the storage claim. $\square$

An optional early exit on $\|Qu - c\alpha\|_2 \leq \eta\sqrt{\|u\|_2^2 + \alpha^2}$ is valid in exact arithmetic and does not increase the worst-case cost. It tests the augmented residual using a preprocessed product. Recomputing the original dense residual $Ax - b$ on every iteration is *not* part of the algorithm; that would reintroduce an avoidable logarithmic cost. A single final original-system check or construction of (26) costs $O(n^2)$ and is harmless.

7 Reproducibility and scope

The accompanying archive contains the mathematical source, a PDF, exact component tests, a floating-point solver prototype, reproducible experiments, and a separate audit of the model and edge cases. These serve different roles.

7.1 Exact component checks

The standard-library test suite uses integers and `fractions.Fraction` for exact checks. All nine test groups pass. They cover injective pattern identifiers, all shifted pattern products, exhaustive small greedy packings,

exact forward and transpose products, comparison-based floors, the uniform parameter inequalities, rational diagonal filtering examples, the $\alpha = 0$ and tiny- α clamping cases, and the error-budget constants. The tests include 147 rational matrix fixtures with both product directions, packing budgets 1 through 6, and integer dimension boundary checks near 16^j for $j \leq 100$. These are implementation checks, not a proof of the universal theorem; the proof is Sections 2–6.

7.2 Floating-point illustrations

The NumPy prototype is explicitly not a finite-precision theorem. It is tested on 15 fixed-seed systems, including ill-conditioned, nonnormal, and one singular inconsistent example. All final original-system checks passed, and all reported errors are below $5\varepsilon/8$. Table 1 gives selected runs; the complete data, seeds, and version metadata are in the archive.

Test family	n	ε	Steps	$\ Ax - b\ _2 / \ x\ _2$
Diagonal, smallest entry 10^{-250}	4	0.25	171	3.1092×10^{-2}
Small singular-value outlier	32	0.10	240	1.4401×10^{-2}
Geometric spectrum	64	0.25	75	4.1896×10^{-2}
Nonnormal bidiagonal	24	0.25	424	3.1829×10^{-2}
Singular, inconsistent (extension)	16	0.25	51	4.4016×10^{-2}

Table 1: Illustrations with augmented-residual early exit. Main experiment runs use dense products for convenience; separate same-draw comparisons test the compressed implementation. The table is not a speed benchmark.

The dense and compressed backends were compared end to end on five of the same draws, with maximum observed relative output difference below 2.0×10^{-16} . Separate product comparisons exercise budgets $k = 2, \dots, 7$, rather than only the small default budgets at modest n . In 300 diagnostic rounding trials the largest observed ratio $\|Q - A\|_2 / (\varepsilon/8)$ was approximately 0.1186. Diagnostic spectral norms and matrix-family construction occur outside the proved algorithm. None of these finite samples establishes its universal success probability.

The solver prototype uses floating-point Gaussians and arithmetic, and uses a Python ceiling for its diagnostic step cap; the paper gives comparison-based implementations for every restricted-model operation. An optional NumPy floor shortcut, optional dense products, and an optional user-imposed iteration cap are clearly marked in the code. A truncated run does not inherit the fixed-time theorem. Exact circuit tests require only the Python standard library; the numerical experiments require NumPy.

7.3 What the result does and does not establish

The answer to the displayed IE-12 cost question is affirmative with an explicit exponent $q = 3$. The success criterion is precisely A -only normwise backward error for the original data. No condition-number bound, input-distribution assumption, rounded-matrix invertibility, or restriction on the direction or magnitude of b is used.

The method still takes $O(\varepsilon^{-2} \log(n+1))$ filter steps. It removes the dimension-dependent logarithm from *total arithmetic cost* through entry-dependent preprocessing, rather than proving a dimension-free black-box iteration rate. Constants are deliberately generous and no optimal exponent in ε is claimed. At modest dimensions the chosen catalogue budget is small, so this construction should not be interpreted as a demonstrated practical improvement over established solvers.

The theorem counts exact-real scalar operations and real-scalar words. It does not bound finite-precision roundoff or the bit lengths of exact intermediates. In particular, a small backward error need not imply a small forward error for an ill-conditioned input. The explicit perturbation (26), not closeness to $A^{-1}b$, is the certificate. The result is presented with a self-contained proof; independent mathematical review would be appropriate before treating it as an externally established resolution of the repository entry.

References

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